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AQA A Level Further Mathematics (7367)

Topic B: Complex Numbers

Sub topic B11: Geometric Problems

Mark Scheme

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AQA A Level Further Mathematics (7367)

Mark Scheme

Topic	**B: Complex Numbers**
Sub topic	**B11: Geometric Problems**

To be used by all students and teachers for AQA A Level Further Mathematics (7367) for exams from 2025 onwards.

Students and teachers of other boards may also find this useful

Mark schemes

Q1.

(a) Centre $-1 - i$ or $(-1, -1)$

B1

Radius 5

ft incorrect centre if used

M1A1F

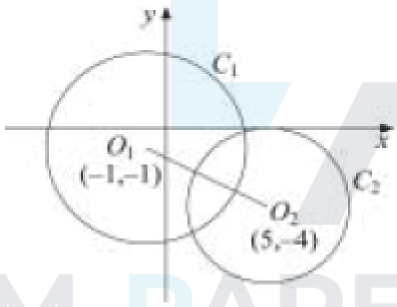
$|z + 1 + i| = 5$ or $|z - (-1 - i)| = 5$

ft $|z + 1 + i| = 10$ *earns M0B1*

A1F

4

(b)



C_1 correct centre, correct radius

ft errors in (a) but fit circles need to intersect and C_1 *enclose (0, 0)*

B1F

C_2 correct centre, correct radius

B1

Touching x-axis

error in plotting centre

B1F

3

(c) $O_1O_2 = 3\sqrt{5}$

allow if circles misplaced but O_1O_2 *is still* $3\sqrt{5}$

M1A1

Correct length identified

m1M1

Length is $9 + 3\sqrt{5}$

ft if r is taken as 10

A1F

5

[12]

Q2.

(a) radius $\sqrt{2}$ centre $-5 + i$

condone (–5, 1) for centre do not accept (–5, i)

B1,B1

2

(b) $\arg(z_1 + 2i) = \arg(-4 + 4i)$

M1

$= \frac{3\pi}{4}$

clearly shown eg $\tan^{-1}\left(-\frac{1}{1}\right)$

A1

2

(c) (i) $|z_1 + 5 - i| = |1 + i| = \sqrt{2}$

B1

1

(ii) Gradient of line from

$(-5, 1)$ to $(-4, 2)$ is $1 \left(\frac{\pi}{4}\right)$

M1 for a complete method

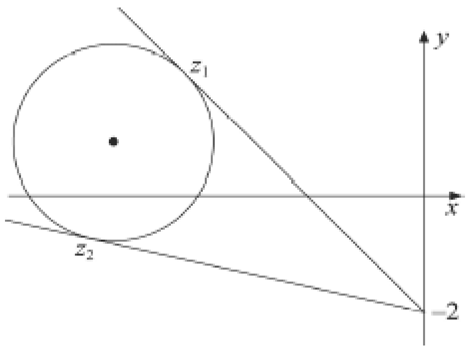
M1A1

radius $\perp$ line $\therefore$ tangent

E1

3

(iii)



Circle correct

ft incorrect centre or radius

B1F

Half line correct

line must touch C generally above the circle

B1

2

(d) z_2 in correct place

B0 if z_2 is directly below the centre of C

B1

with tangent shown

B1

2

[12]

Q3.

(a) Explanation

E1 for $i = e^{\frac{\pi i}{2}}$ *or* $iz_1 = -y_1 + ix_1$

E2,1,0

2

(b) (i) Perpendicular bisector of AB

B1

through O

B1

2

(ii) half-line

If L_2 is taken to be the line AB give B0

B1

from B

B1

parallel to OA

B1

3

(c) $(1 + i)z_1$

ft if L_2 taken as line AB

M1A1

2

[9]

Q4.

(a) $\frac{1+i}{1-i} = \frac{(1+i)^2}{1-i^2} = i$

AG

M1A1

2

(b) $|z_2| = \sqrt{\frac{1}{4} + \frac{3}{4}} = 1 = |z_1|$

M1A1

2

(c) $r = 1$

PI

B1

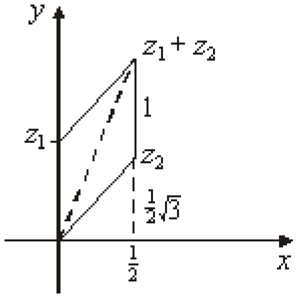
$\theta = \frac{1}{2}\pi, \frac{1}{3}\pi$

Deduct 1 mark if extra solutions

B1B1

3

(d)



Positions of the 3 points, relative to each other, must be approximately correct

B2,1F

2

(e) Arg(z_1+z_2) = $\frac{5}{12}\pi$

***Clearly** shown*

B1

$\tan\frac{5}{12}\pi = \dfrac{1+\frac{1}{2}\sqrt{3}}{\frac{1}{2}}$

Allow if B0 earned

M1

$= 2 + \sqrt{3}$

AG must earn B0 for this

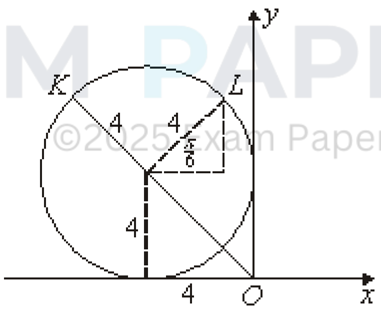
B1

3

[12]

Q5.

(a)



Circle

B1

Correct centre

B1

Touching both axes

B1

3

(b) |z| max = OK

Accept $\sqrt{4^2 + 4^4}$ *+ 4 as a method*

M1

$= \sqrt{4^2 + 4^4} + 4$

Follow through circle in incorrect position

A1F

$= 4(\sqrt{2} + 1)$

AG

A1F

3

(c) Correct position of z , ie L

M1

$a = -\left(4 - 4\cos\frac{1}{6}\pi\right)$

$= -\left(4 - 2\sqrt{3}\right)$

Follow through circle in incorrect position

A1F

$b = 4 + 4\sin\frac{1}{6}\pi = 6$

A1F

3

[9]

Examiner reports

Q1.

The coordinates of the centre of the circle in part (a) were usually obtained, but the notation was often poor and it was not uncommon to see the centre of the circle C_1 written as (–1, –i) and, on the diagram, the scale on the y-axis written as i, 2i, 3i and so on. Also, radius and diameter were commonly confused.

Sketches in part (b) varied considerably, the best being those who used compasses for their circles. These were generally readable with centre and radius indicated. However, some candidates chose to draw their circles by plotting points and joining up by freehand. These sketches turned out to be very poor. The circle C_2 was sometimes mistakenly drawn in the incorrect quadrant through choice of centre as (–5, 4) whilst others either failed to realise that the circle C_2 touched the x-axis or drew a circle touching both axes.

In part (c), although many candidates placed z_1 and z_2 on their diagram in the approximately correct positions, not all realised that these points were at the intersections of C_1 and C_2 and the line O_1O_2, and even when they did, the finding of the length O_1O_2 proved to be beyond many.

Q2.

Apart from the odd sign error, part (a) was done correctly. Answers to part (b) and to some extent part (c)(i) lacked detail especially as the results were given or implied. Very few candidates scored credit in part (c)(ii) and when it was attempted it was almost always solved by finding the cartesian equation of the circle C and the line L and by showing that the resulting quadratic equation giving the points of intersection of C and L had two coincident roots. This method, of course, involved a considerable amount of algebraic manipulation, when in fact the consideration of the gradient of the line joining the centre of C to the point represented by z_1 would have given the result immediately. However, this was very rarely seen. It was surprising to see many candidates draw the circle C and the line L intersecting in two points on the Argand diagram (part (c)(iii)) in spite of the results candidates had been requested to show in the earlier parts of the question. There were many incorrect solutions to part (d), the most common being to place z_2 vertically below the centre of C.

Q3.

Explanations in part (a) were very unclear and generally far from convincing. Candidates generally referred to what had happened to the coordinates of the points represented by z_1 and z_2, but few made allusion to the significance of i in the iz_1. The neatest solutions came from candidates who considered multiplication of a complex number by i as a rotation anticlockwise of $1/2\pi$. Inaccurate copying of the diagram in part (b) caused loss of marks. For instance, although candidates knew that the locus L_1 was the perpendicular bisector of AB, poor diagrams meant that their line did not pass through the origin. Again, for the locus L_2, although the majority of candidates drew a half line through B, their line was not always parallel to OA. Part (c) proved to be beyond most candidates probably because few realised that the point of intersection of $L_1$and L was, in fact, the fourth vertex of the square whose three other vertices were A, O and B

Q4.

Part (a) was well done, as was part (b). In part (c) it was surprising to note how many candidates could not express the complex number in the form $re^{i\theta}$, although z_2 was almost invariably correctly written as $e^{i\frac{\pi}{3}}$ Errors in part (c) however did not deter candidates from drawing a correct Argand diagram as they usually used the form $\boldsymbol{a}$ + i$\boldsymbol{b}$ when plotting their points. Although the vast majority of scripts ended with tan $\frac{5\pi}{12}$ = 2 + √3 , very, very few candidates gave convincing proof that arg (z_1+z_2) was $\frac{5\pi}{12}$, but rather seemed to take it for granted.

Q5.

Most candidates realised that the locus in part (a) was a circle although it was frequently drawn in an incorrect quadrant, and occasionally with a radius of 2 rather than a radius of 4. The correct answer to part (b) was usually obtained although sometimes with a less than convincing argument. There were relatively fewer correct solutions to part (c). Those candidates who addressed the geometry of the figure were the most successful, but those who converted the equations of the circle and line into the Cartesian form and then attempted to solve a pair of simultaneous equations usually abandoned their solution after making algebraic errors.